



SGL3: Pathology of: 1. Intestinal Neoplasms 2. Diverticulum

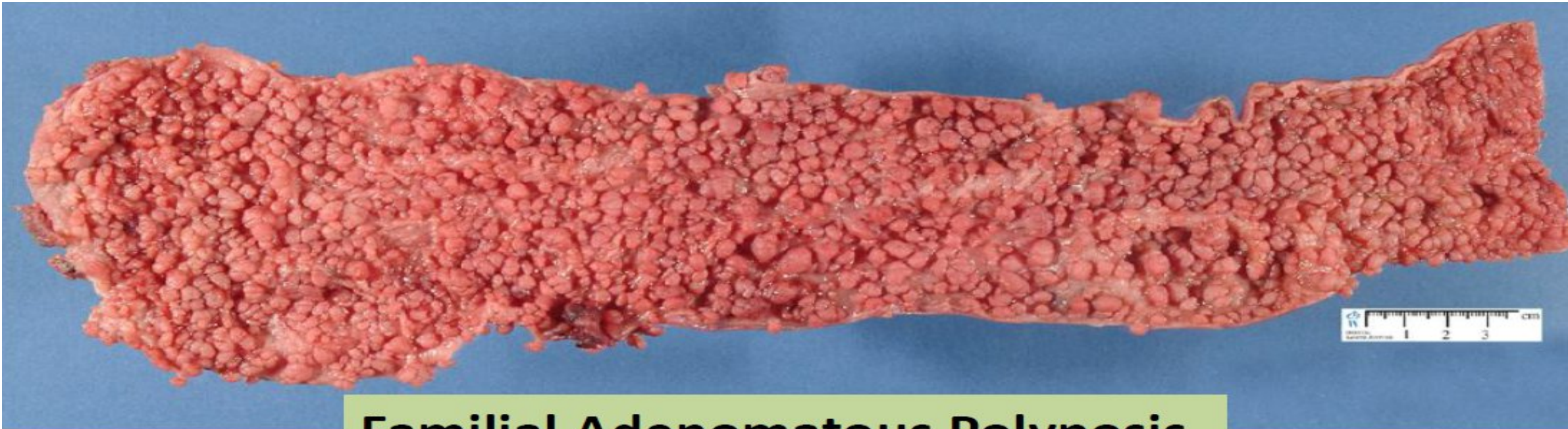
**By: Consultant Pathologist
Asst. Prof. Dr. Jawher Tahir Omer
MBChB / FIBMS-Pathology**

(Intestinal Neoplasms)

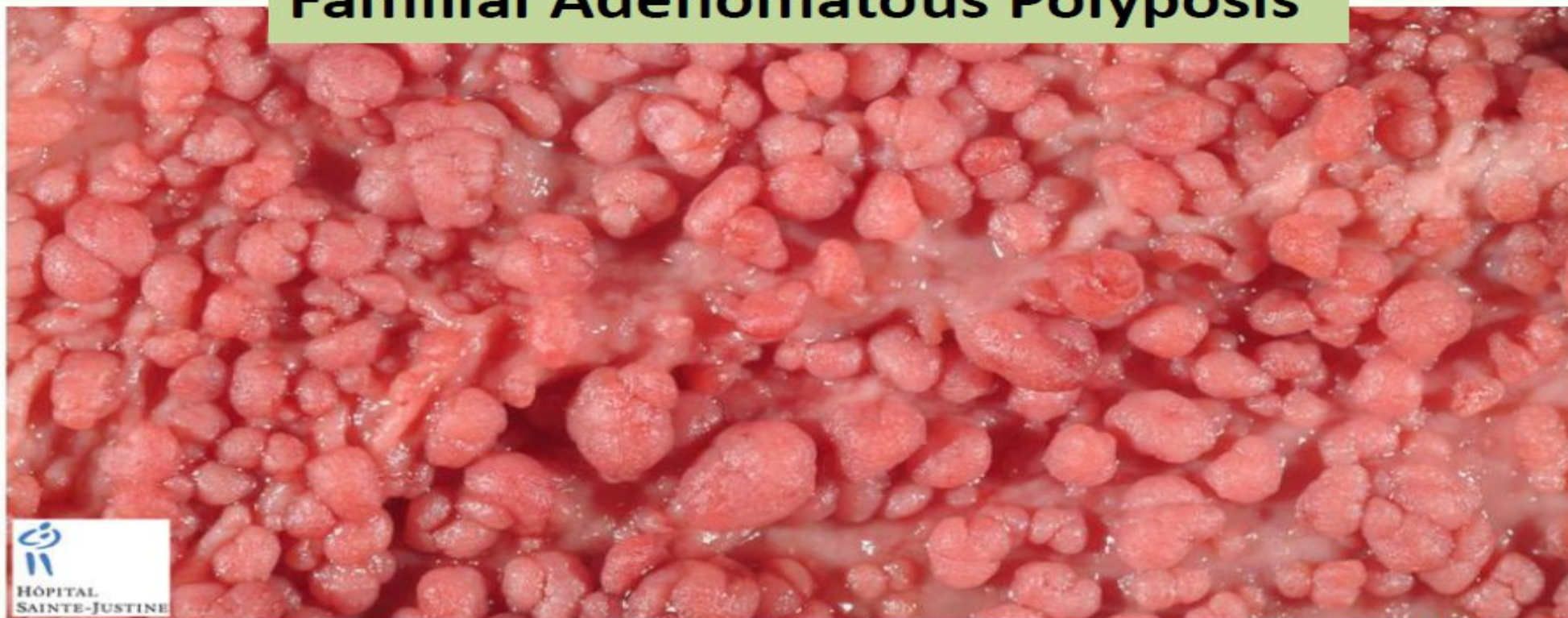
Familial Adenomatous Polyposis (FAP):

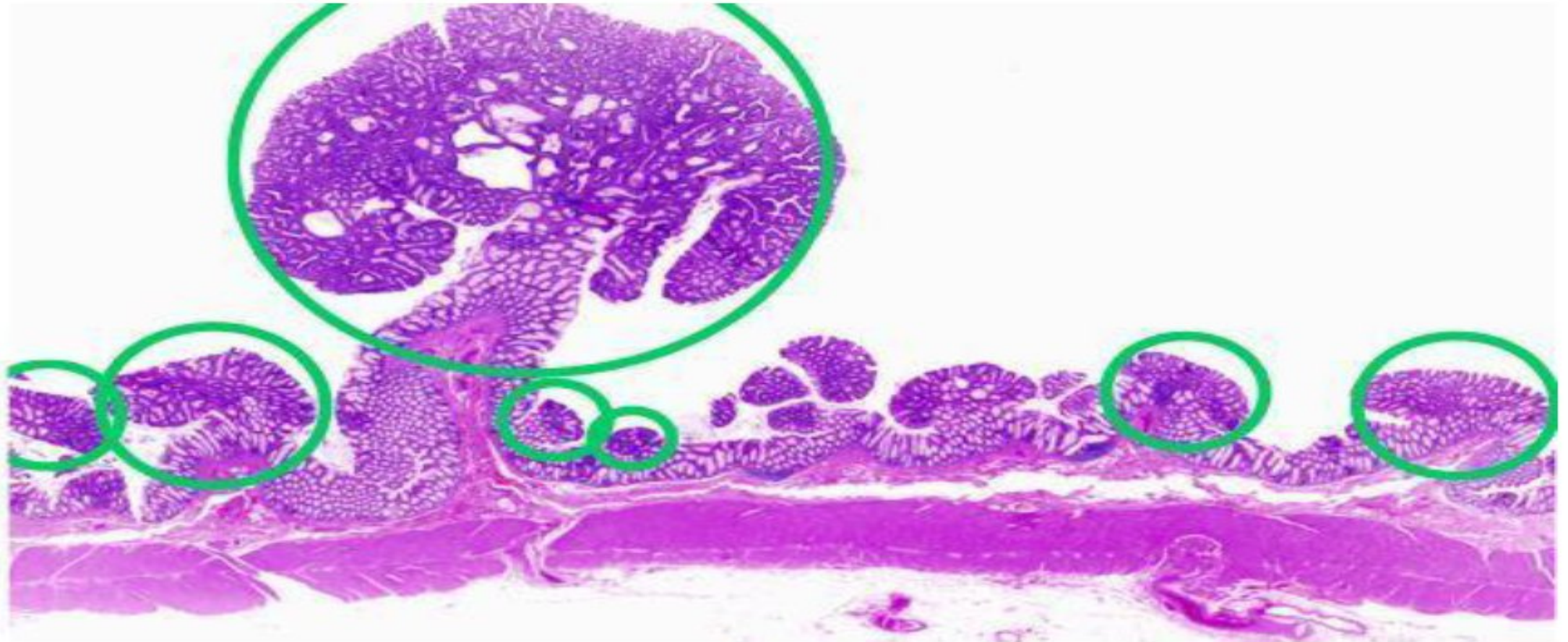
- AD disorder, caused by mutation of Adenomatous Polyposis Coli (APC) gene on long arm of chromosome 5.
- APC is a tumor suppressor gene and its products (WNT signaling) regulates cell proliferation.
- Marked by the appearance of numerous colorectal adenomas by the teenage years.
- A count of at least 100 polyps is necessary for a diagnosis of classic FAP, and as many as several thousand may be present.

- Colorectal adenocarcinoma develops in 100% of patients with untreated FAP, often before the age of 30 years.
- Prophylactic colectomy in late 2nd decade is standard therapy for persons carrying APC mutations.
- However, patients remain at risk for other extracolonic manifestations, including neoplasia at other sites.



Familial Adenomatous Polyposis





Familial Adenomatous Polyposis

Morphologically indistinguishable from sporadic adenomas.

Intestinal malignancy:

- Adenocarcinoma of LI is the commonest GIT malignancy.
- Adenocarcinoma of SI is very uncommon in comparison with LI.
- An exception is the area of duodenum around the ampulla of Vater, where adenomatous lesions are well recognized.
- Adenomas and adenocarcinomas of SI and LI have similar microscopic features.

Colorectal carcinoma (CRC):

- The vast majority are adenocarcinoma.
- CRC incidence peaks at 60 to 70 years of age.
- Males are affected slightly more often than females.
- Most CRC arise in adenomatous polyps.

▪ Dietary factors:

- **Low intake of unabsorbable vegetable fiber and high intake of refined carbohydrates:** possibly due to decreased stool bulk and altered composition of intestinal microbiota which may increase synthesis of potentially toxic oxidative by-products of bacterial metabolism.
- **High intake of fat:** enhances hepatic synthesis of cholesterol and bile acids which can be converted into carcinogens by intestinal bacteria.
- **Deficiencies of vitamins A, C, and E:** antioxidants effect.

- Several studies suggest that aspirin or other NSAIDs have a protective effect:-
 - This effect is possibly mediated by inhibition of COX-2 enzyme,,
 - COX-2 enzyme is necessary for production of prostaglandin E2 that promotes epithelial proliferation.

Pathological and clinical features

Right colon

- Polypoid fungating mass.
- Present as insidious anaemia.



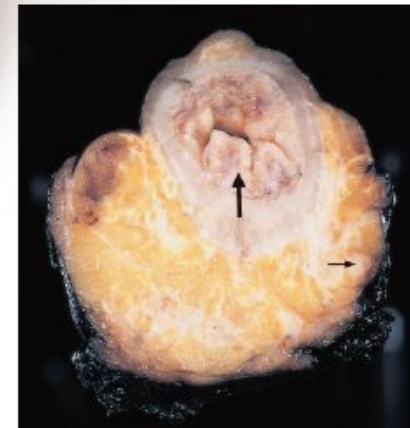
Left colon

- Circumferential infiltrated 'napkin-ring' constriction.
- Present as:
 - Bleeding PR
 - Change bowel habit
 - intestinal obstruction

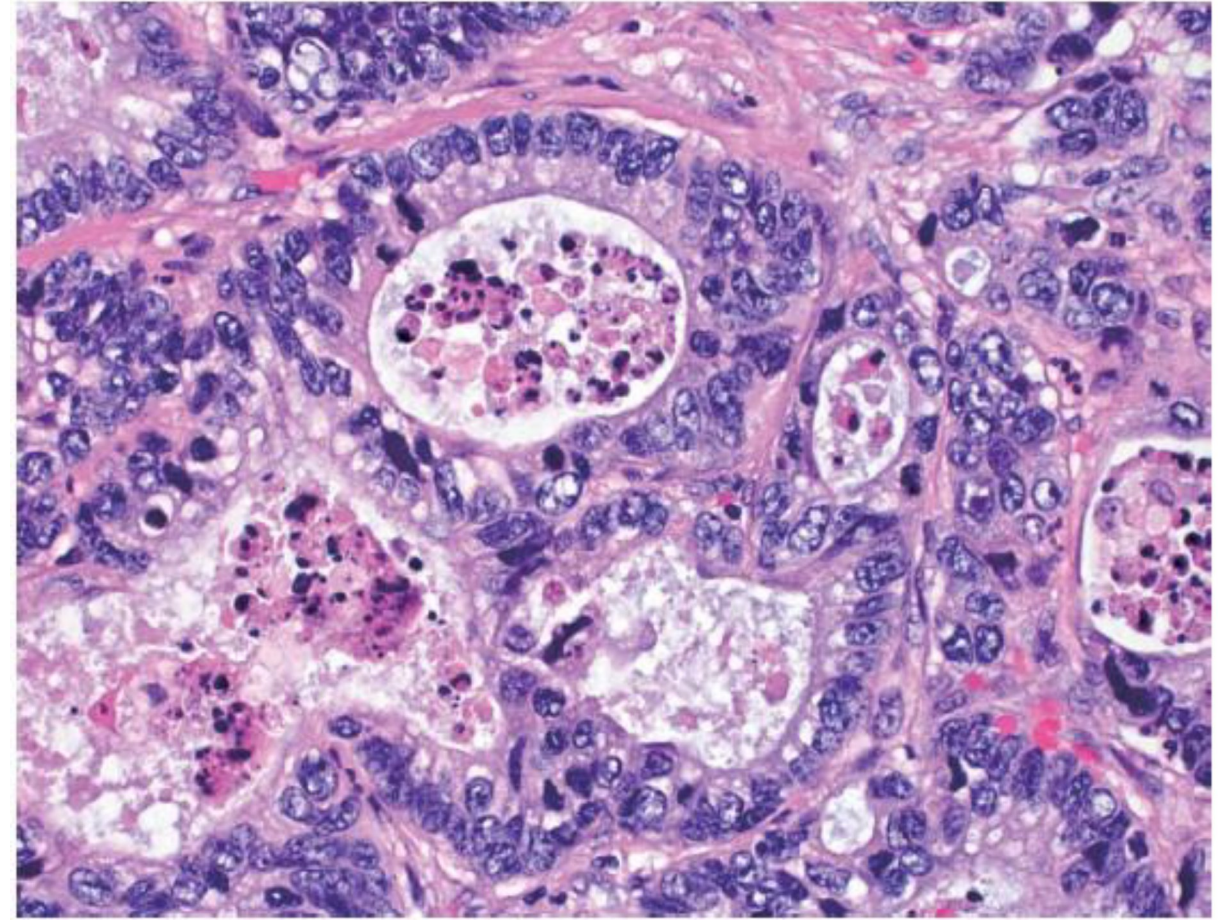
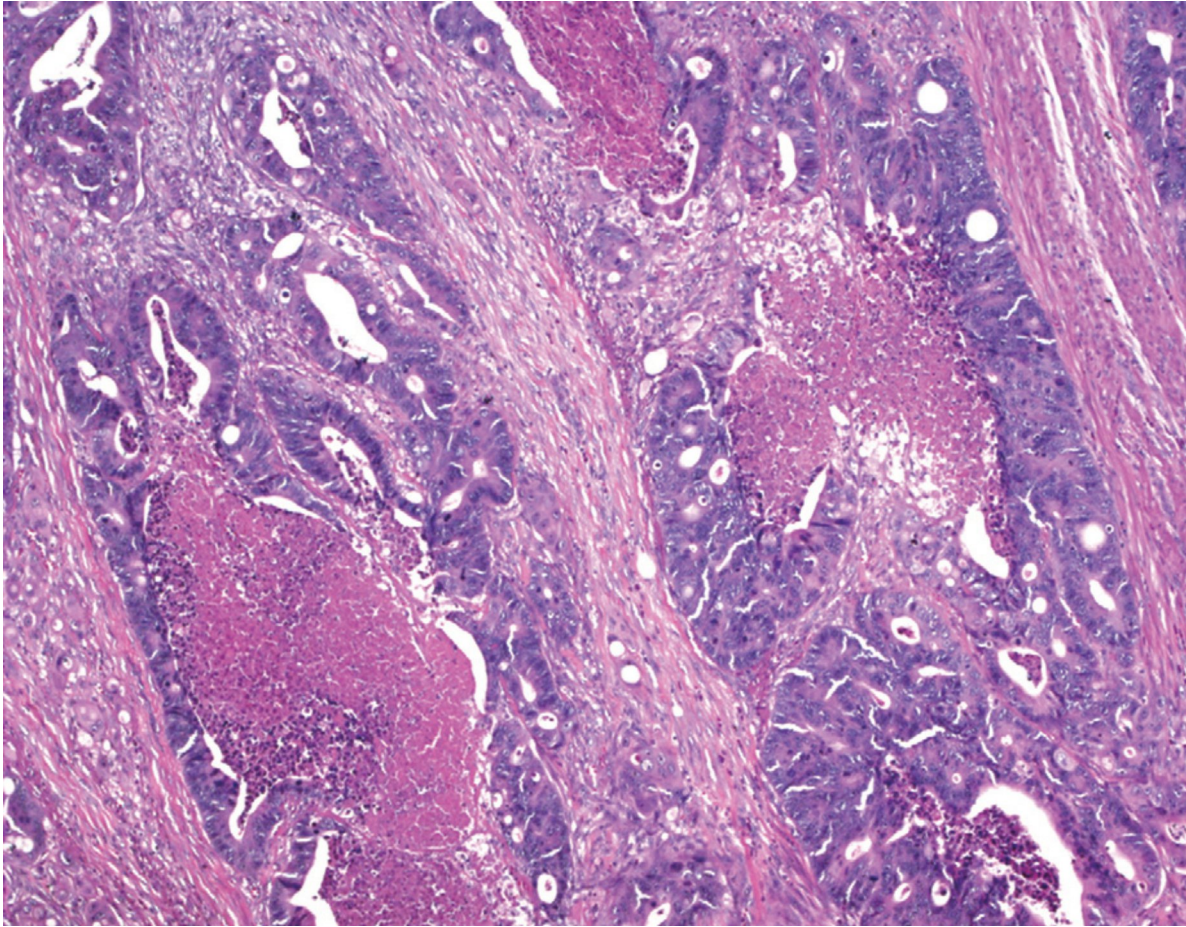


Rectal tumours

- Ulcerated raised plaques
- Present as:
 - Fresh bleeding PR
 - Tenesmus
 - Obstructions



Adenocarcinoma (Histology): Malignant cells arranged in glandular pattern and showed marked necrosis with infiltration of the muscle layer.



▪ Spread of CRC:-

- **Direct spread:** through the intestinal wall.
- **Lymphatic spread:** through lymphatics into regional LNs.
- **Blood spread:** through blood-borne metastases in the liver (through portal vein) and through systemic veins to lungs, bone, brain,,,etc....
- **Transcoelomic spread:** through the peritoneal cavity.

NeuroEndocrine Tumor (NET):

- Arise from the diffuse components of endocrine system (neural crest embryologic derivatives).
- The most common site of origin of such tumors is in the midgut-derived epithelium of ileum and appendix .
- Usually called carcinoids “carcinoma-like” in practice, but the term ‘neuroendocrine cell tumor’ is now preferred.

- As a rule, these neoplasms all have malignant potential but many are very indolent.
- ileal tumors are of low-grade malignancy, and not infrequently metastasize to local nodes or liver.
- Appendiceal NET tend to behave in benign manner.

- **Clinical presentations:**

- **Local mass effect:** it may lead to complications like intestinal obstruction and intussusception.
- **Systemic effects:** these presentations are due to the wide range of peptides produced by the tumor cells and may be active either locally or systemically.

▪ **NET systemic effects:**

- Foregut NET (including pancreatic) may produce hormones such as gastrin or insulin.
- Midgut NET more often produce smaller active products such as serotonin [5-hydroxytryptamine (5-HT)], this is passed in portal circulation to liver where it is inactivated to 5-hydroxyindoleacetic acid (5-HIAA).
- Elevated 5-HIAA levels in urine can be used as an aid in diagnosing these tumors.

▪ NET systemic effects....Continue:

○ Carcinoid syndrome:

- A systemic disorder characterized by flushing attacks, diarrhea, and occasionally endothelial thickening of the right side of the heart leading to tricuspid and pulmonary stenosis.
- It is usually seen when intestinal NET have metastasized to liver, and 5-HT (serotonin) is released directly into the systemic circulation.

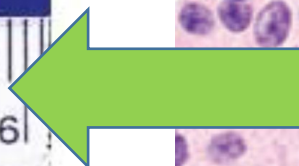
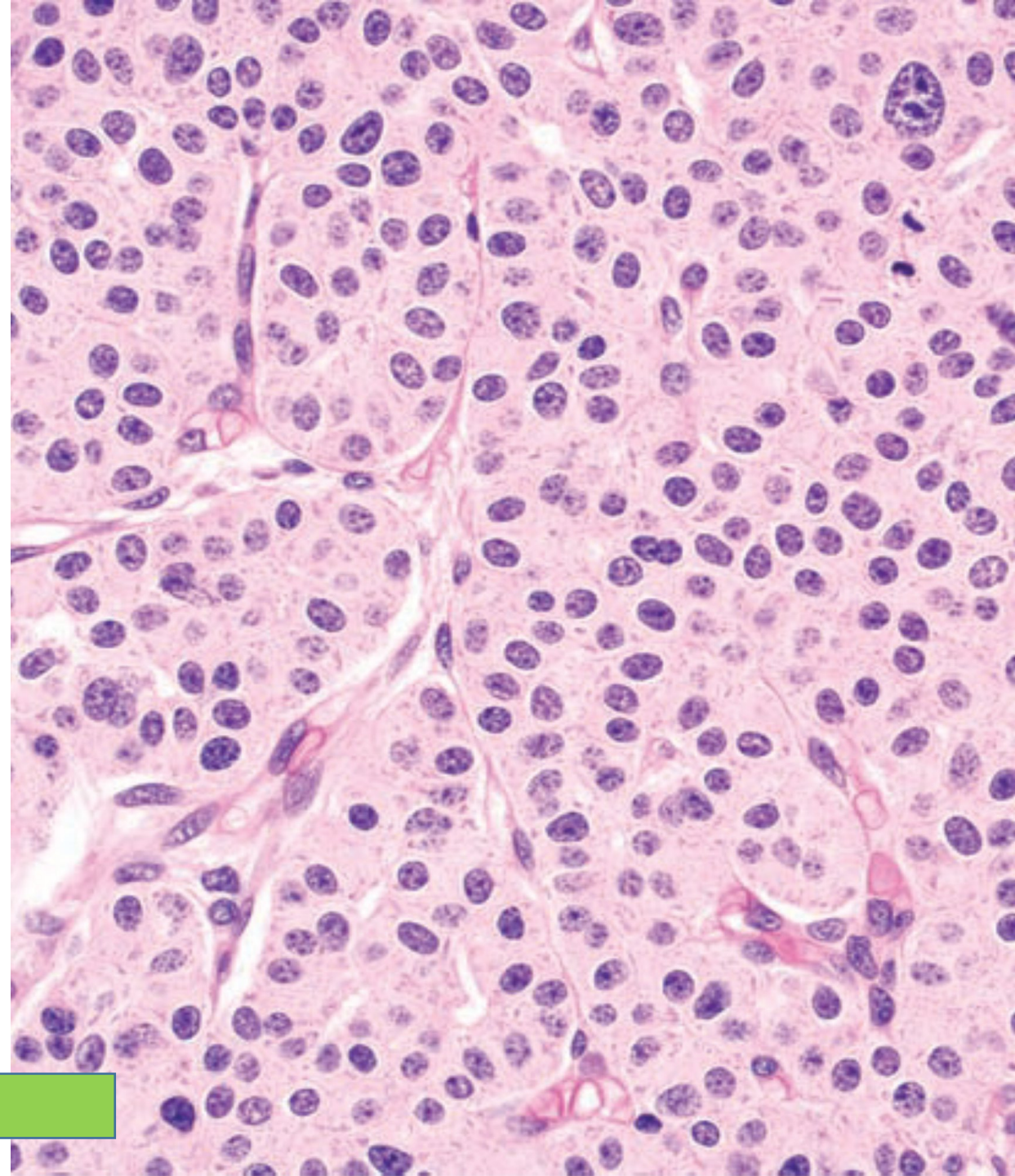
Pathology of NET:

- **Grossly:**

- Intramural or submucosal masses that create small polypoid lesions.
- Tend to be yellow or tan in color and very firm.

- **Histologically:**

- NET are graded depending on their degree of differentiation (atypia), and the mitotic activity into: Grade 1, 2 and 3.
- The most differentiated form is composed of islands, trabeculae, strands, glands, or sheets of uniform cells with scant pink granular cytoplasm, and a round to oval stippled nucleus.



Gastrointestinal Stromal Tumors (GISTs)

- The most common mesenchymal tumor of GIT.
- Believed to arise from the interstitial cells of Cajal which act as “intestinal pacemaker cells” as they coordinate peristalsis.
- May occur at any part of GIT, most common in stomach (60% of tumors) followed by the small bowel (30%).

Presentation and behavior of GISTs:

▪ Presentation:

- Depend on location and size of the tumor.
- Most common presentations are nonspecific:- pain, GI bleeding, mass effect, and obstruction.

▪ Behavior:

- The most important parameters for predicting the behavior of GISTs are:- size, anatomic location, and mitotic rate.
- Accordingly, the tumors are classified as very low risk, low risk, intermediate risk, or high risks.
- Higher risk tumors has more probability for recurrence and/or metastasis.

Pathology of GIST:

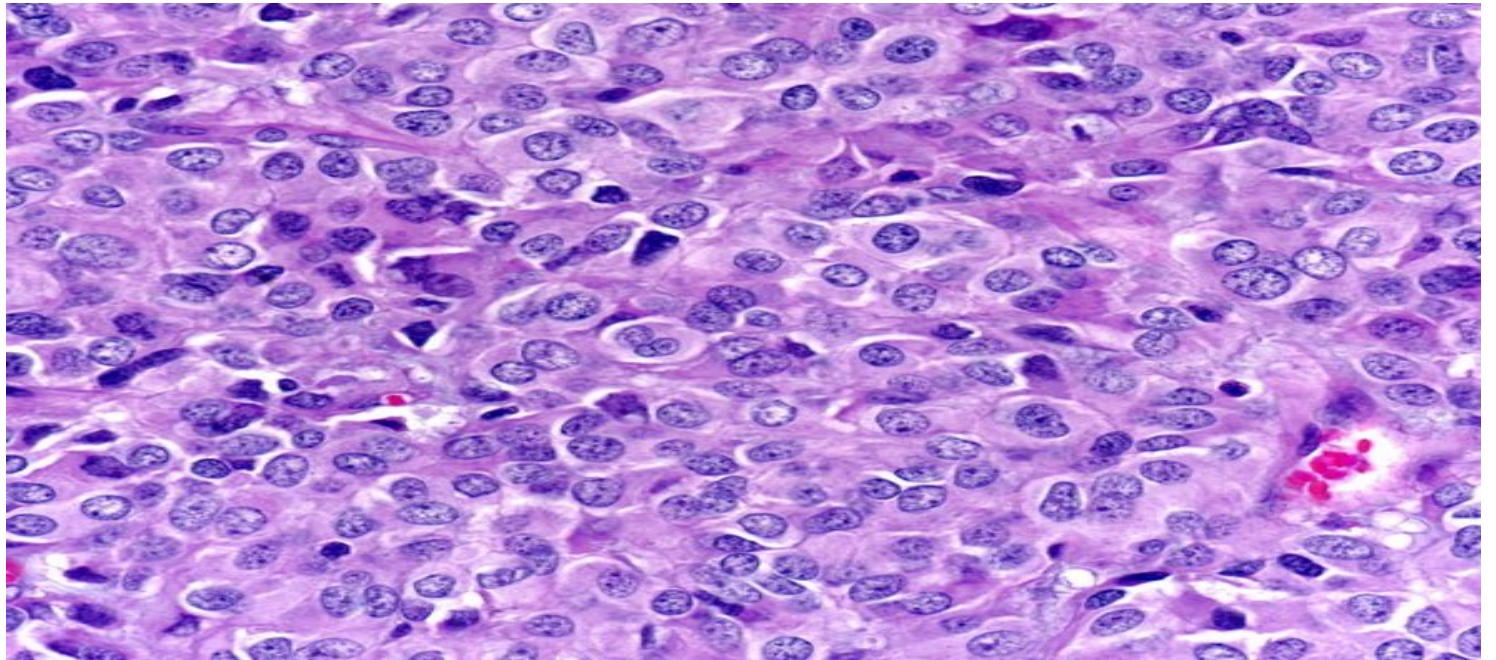
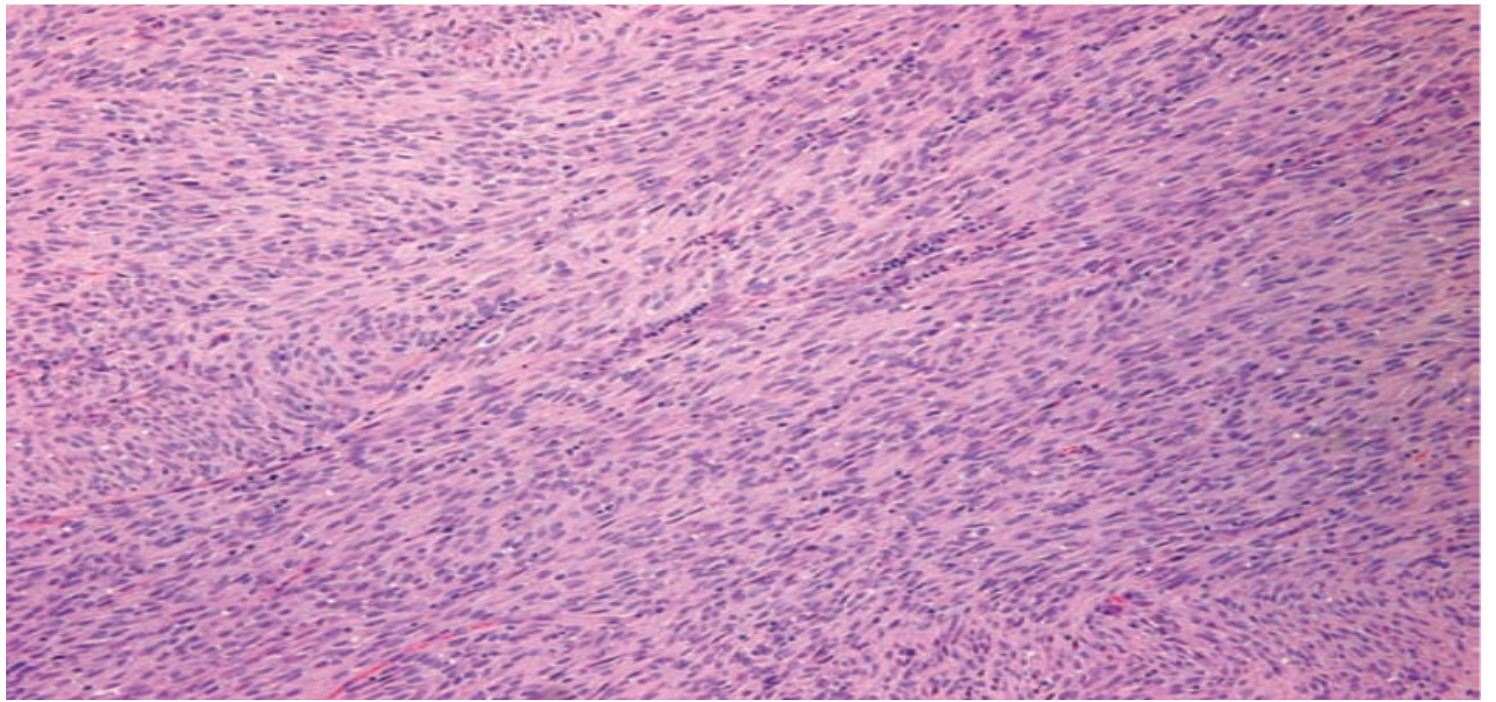
- **Gross pathology:**

- Usually well circumscribed mass(es).
- Some times polypoidal morphology.

- **Microscopic pathology:**

- The majority (70%) of GISTs are spindle cell lesions.
- About 20% have epithelioid morphology.
- 10% are mixed Spindle and epithelioid.

GIST Pathology: gross & microscope



Diverticulum:

- A diverticulum is a blind outpouching of the mucosa in, or through the muscularis propria.
- May be congenital or acquired:-
 - Congenital divert.: are true divert. i.e. lined by all three layers of the gut wall.
 - Acquired divert.: either lack muscularis propria entirely, or have an attenuated muscularis propria.

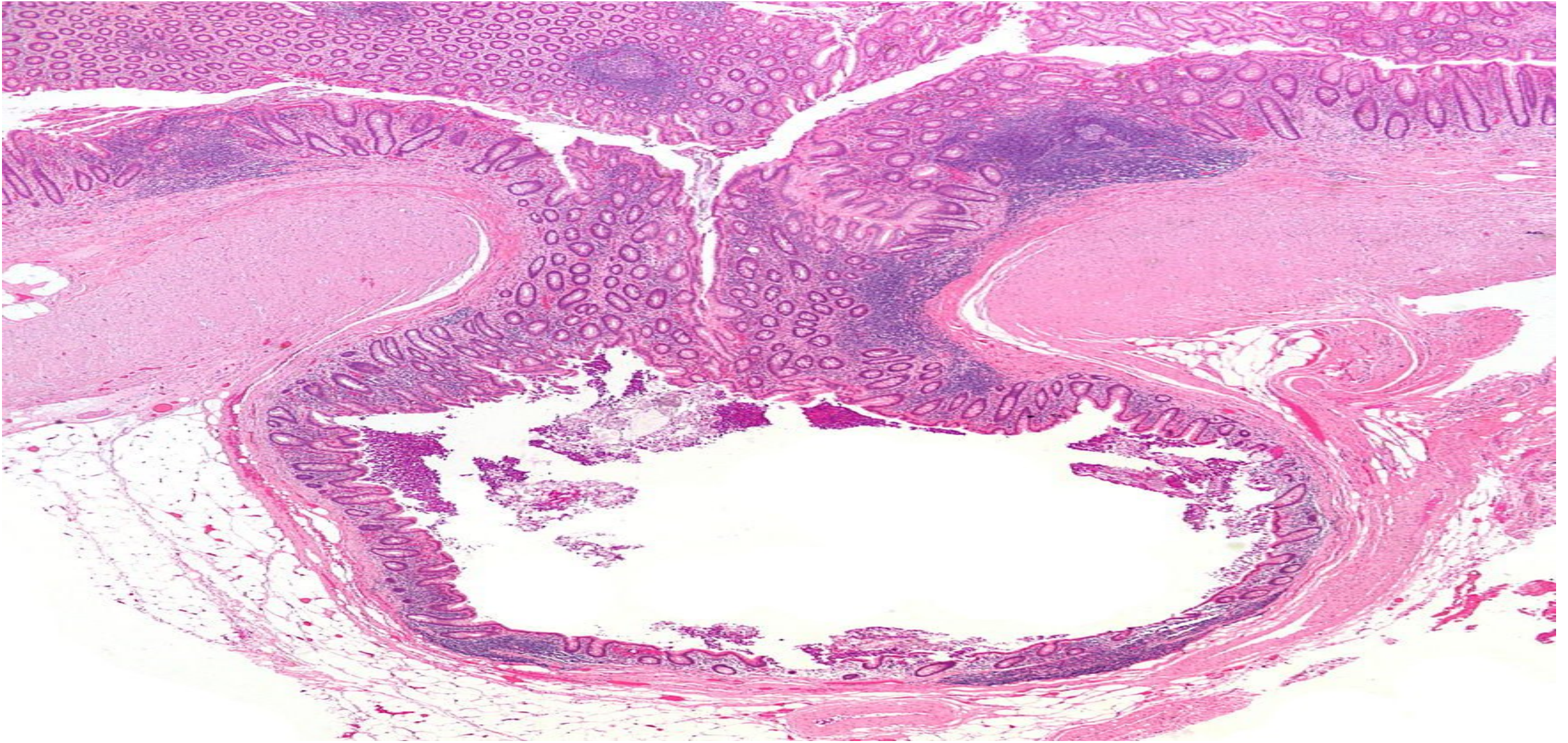
- Diverticulum may develop in any part of GIT.
- The commonest true congenital diverticulum is **Meckel diverticulum.**
- The most common site of **Acquired diverticula is sigmoid colon.**

Acquired diverticulum (Sigmoid colon)

- Acquired diverticulum develops in the presence of increased intraluminal pressure through weak points in the intestinal wall.
- Increased intraluminal pressure is seen in low fibers diet.
- The weak points of colon are where the inner circular muscularis propria is penetrated by nerves, arterial vasa recta and their connective tissue sheaths.



The sigmoid colon at the right appears lighter in color than the adjacent small intestine and has a band of taenia coli muscle running longitudinally. Protruding from the sigmoid colon are multiple rounded bluish-gray diverticula.



Sigmoid colon /diverticulum: mucosa/submucosa invaginate into the muscularis propria

- **Clinical features:**

- Asymptomatic
- Abdominal pain
- Constipation &/or diarrhea

- **Complications:**

- Inflammation (diverticulitis) & abscess.
- Intestinal obstruction (by scarring)
- Fistula formation
- Perforation & peritonitis
- Bleeding

Meckel diverticulum:

- Occurs as a result of failure of involution of vitelline duct which connect the lumen of developing gut to the yolk sac.
- It extends from the antimesenteric site.
- The mucosal lining may resemble that of normal small intestine, or contain ectopic pancreatic or gastric tissue which may result in inflammation, ulceration& bleeding, and obstruction.

- The rule of 2's in Meckel diverticulum:
 - Occur in nearly 2% of population.
 - Located in ileum and are generally presents within 2 feet (60 cm) of ileocecal valve.
 - Approximately 2 inches (5 cm) long.
 - Twice as common in male.
 - Most often symptomatic by age of 2 years.
 - 2/3rd have ectopic tissue (gastric or pancreatic).
 - 2% become symptomatic.

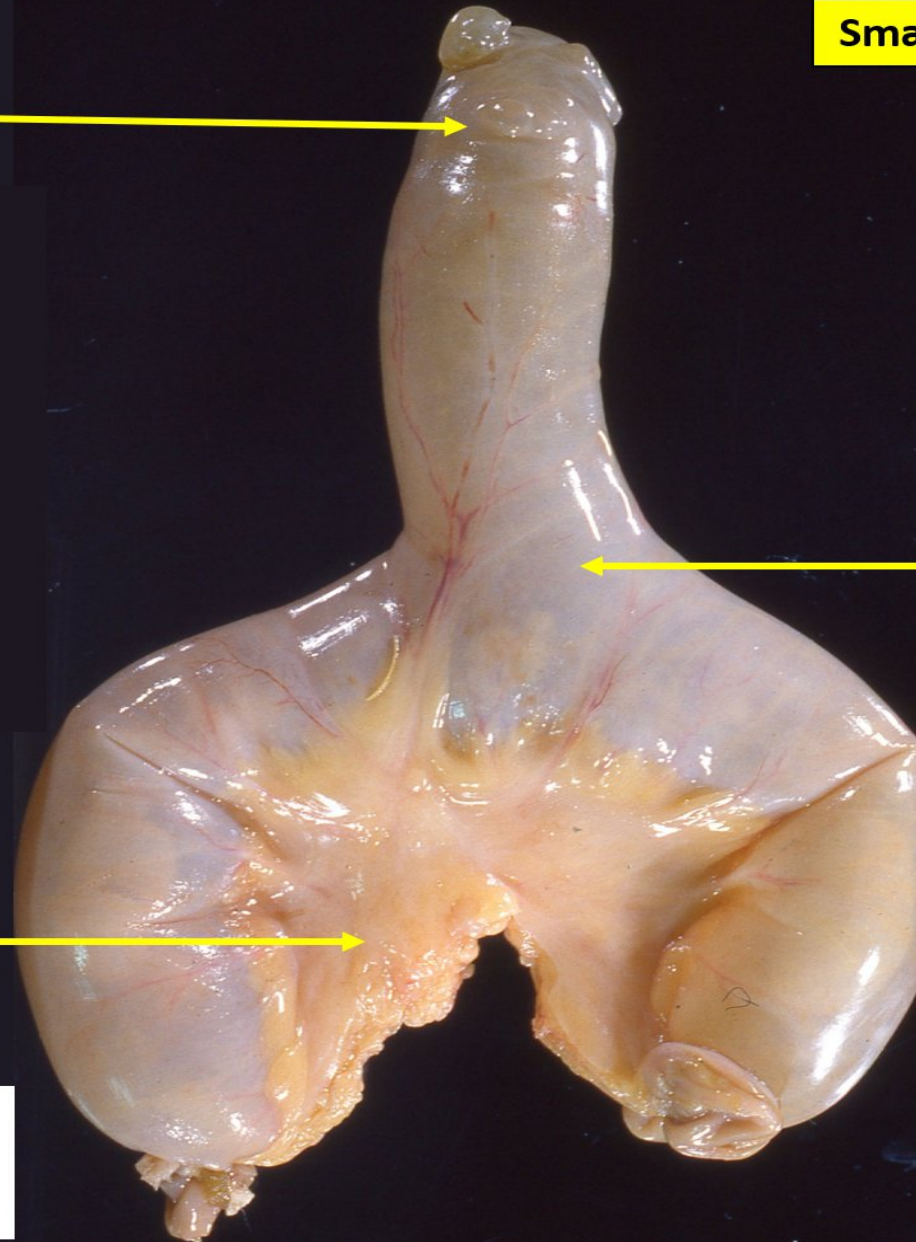
Small Intestine – Meckel's diverticulum

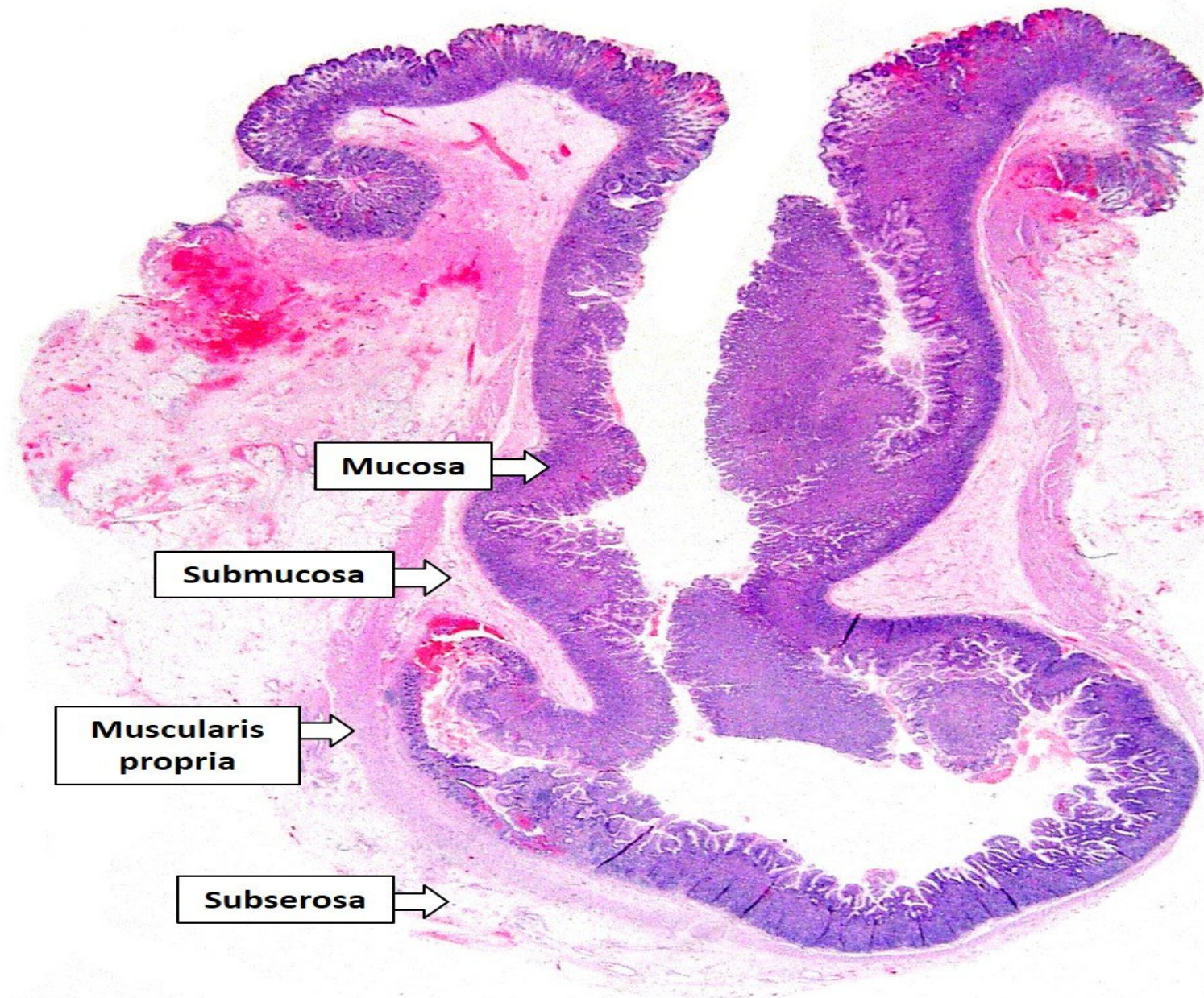
Blind ended tip of
diverticulum

Base of diverticulum is
connected to small bowel

Small bowel mesentery

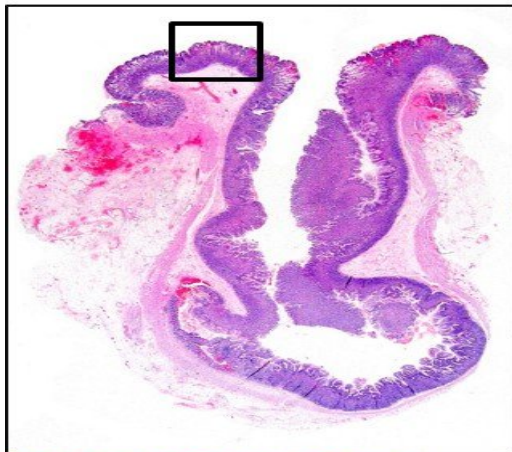
@Pathweb





Small bowel: Meckel's diverticulum

A true diverticulum with outpouching of all four layers (mucosa, submucosa, muscularis propria and serosa)



Small bowel: Meckel's diverticulum (medium power)

Left: Small bowel mucosa

Right: Heterotopic gastric mucosa (this can be seen in Meckel's diverticulum and predispose to peptic ulcer)

